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Global seroprevalence of *Toxoplasma gondii* in pregnant women: a systematic review and meta-analysis

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Abstract

Background Toxoplasmosis in pregnancy is associated with serious and irreversible maternal and fetal detrimental consequences. Also, different seroprevalence of *Toxoplasma gondii* in pregnancy is reported in many countries. The present systematic review and meta-analysis study aimed to determine the global seroprevalence of *Toxoplasma gondii* in pregnant women.

Methods This study was conducted based on the PRISMA 2020 criteria. Initial searching was conducted using MeSH (Medical Subject Headings)-based keywords with no time limitation (by August 1, 2024). Collected papers were transferred to Citation Management Software (EndNote). Duplicate studies were merged and primary and secondary screenings were applied based on the inclusion/exclusion criteria. Validation was considered to find high-quality assessments. Finally, eligible extractable papers were enrolled for data collection. Data was analyzed using Comprehensive Meta-Analysis software (v.2). The random effects model was used in case of I^2 index above 50%. In order to investigate the factors affecting the heterogeneity of studies, meta-regression tests were used to examine factors such as sample size and year of study.

Results One hundred thirty-eight eligible studies with a total sample size of 135,098 pregnant women individuals were selected for data extraction and analysis. The heterogeneity index was found high (I^2 :98.9) and the random effect model was used for analysis. The Egger test revealed the absence of publication bias in collected studies (p :0.088). Thus, the global seroprevalence of *Toxoplasma gondii* in pregnant women was reported at 36.6% (95% CI:33.7–39.6). The highest prevalence reported based on meta-analysis was reported in South America with 52.8% (95% CI:46.6–59), while only 15 studies were reviewed in this continent, most of which were in Brazil. Therefore, after the continent, the highest prevalence reported was reported in Africa with 46.8% (95% CI:39.5–54.3). Also, the lowest prevalence reported based on meta-analysis was in North America with 19.7% (95% CI:8.4–39.6) and Europe with 24.6% (95% CI:17.8–32.9).

Conclusion This study revealed a high level of seroprevalence of *Toxoplasma gondii* in pregnant women worldwide. This value mostly depends on the individual's age, lifestyle, and disease awareness regarding toxoplasmosis in pregnant women. Thus, public awareness, along with comprehensive health programs regarding the detrimental effects of toxoplasmosis in pregnant women, seems necessary for prevention or even early diagnosis of toxoplasmosis in pregnant women.

Keywords *Toxoplasma gondii*, Pregnant women, Pregnancy, Meta-analysis

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Background

Toxoplasmosis is an infectious-parasitic disease in humans and animals, caused by *Toxoplasma gondii* (TG) [1]. Felines are the only definitive hosts of this parasite and other vertebrates are considered intermediate hosts [2]. TG infection is transmitted through the presence of oocysts in water and food or soil (contaminated with cat feces), raw meat, organ transplantation, blood transfusion, and also placenta to fetus transfer during pregnancy [3–5].

Toxoplasmosis can affect healthy and immunocompromised individuals and can also be a life-threatening disease in immunosuppressed individuals [3–5]. The parasite is transmitted through raw meat containing TG cysts or water containing oocysts from cat feces. Although felines are the only definitive host, TG can infect and replicate in almost any nucleated vertebrate cell [4, 5]. The life cycle TG has an asexual phase that occurs in nucleated cells and a sexual phase in the digestive tract of cats. Fertilized gametes from sexual reproduction are produced in the cat's small intestine, and excreted oocysts can persist in the environment for up to 18 months. Both waterborne and foodborne transmission have been reported [5].

This pathology is demonstrated in three forms of acute, latent, or cutaneous. Although TG infection is often asymptomatic, the acute type of TG infection is resembling the Influenza symptoms such as fever, sore throat, weakness, inflamed lymph nodes, and muscle pain [6–9]. However, Toxoplasmosis is more severe in immunocompromised individuals with detrimental effects on the fetus [10].

Toxoplasmosis is often self-limiting in people with healthy immune systems, but it is a significant risk in immunocompromised individuals and pregnant women [4, 11]. Acute TP is associated with many complications for the fetus and pregnant mother [12, 13]. The risk of vertical transmission of TG from the placenta to the fetus increases from 25% in the first to 65% in the third trimesters followed by more severe complications such as miscarriage, premature birth, congenital-related Toxoplasmosis, retinopathy, neuropathy, homeopathy, hydrocephaly, microcephaly, and cerebral calcification. Thus, early diagnosis and treatment of acute Toxoplasmosis in pregnant women are necessary to reduce the severity of complications [3, 4, 12–20].

The TG seroprevalence of women in reproductive age or pregnant cases ranges from 1.4% to 85%, globally [18–20]. If this infection occurs during pregnancy and infects the pregnant mother, it can lead to congenital toxoplasmosis, which can be a threat to the health of the baby. Clinical manifestations of the infection can be observed later in infancy, childhood, or adolescence.

Epidemiological studies of this disease show that from 1988 to 1994, the seroprevalence of infected women aged 15 to 44 years was 15%, while the prevalence decreased to 11% in 1999–2004, and in 2009–2010, the seroprevalence was reported to be 9.1% [19]. The highest TG seroprevalence is related to tropical regions, lower altitudes areas, developing (South America and Africa), and Middle East countries [4, 21]. However, high seroprevalence of TG is reported in many developed countries (like France) where people consume uncooked meat [21]. These differences indicate the alteration in reported cases of TG seroprevalence in different regions [21].

Serological screening is a valid procedure for diagnosis of acute (or chronic TG infection [16, 17]. After acute infection diagnosis in a pregnant mother, some antibiotics such as Spiramycin are prescribed with an effectiveness rate of >60% to inhibit fetal congenital infection [7, 18, 19].

Assessment of the scientific dimensions of *Toxoplasma gondii* in pregnant women is important due to the development of preventive measures, early diagnosis, and treatment of acute Toxoplasmosis. Given that different reports of the prevalence of this disease in pregnant women around the world report different prevalences and high heterogeneity, Thus, this systematic review and meta-analysis study aimed to report the Global seroprevalence of *Toxoplasma gondii* in pregnant women.

Methods

This study was conducted based on the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) statement criteria (2020) [22]. The initial search was conducted on August 1, 2024, and updated on August 28, 2024 with no time limitation. The searching strategy was designed using the MeSH-based keywords of “Prevalence”, “Factors”, “Outcome”, “*Toxoplasma gondii*”, “Pregnant women”, and “Pregnancy” in major electronic databases, including PubMed, ScienceDirect, Scopus, Web of Sciences (WoS), Embase, and G.Scholar. References of included papers were also reviewed to cover the maximum number of eligible studies. All papers reporting the seroprevalence of *Toxoplasma gondii* in pregnant women were included in this study. Also, other variables (e.g., geographic location, diagnostic method, Age, Sample size) were collected for comprehensive assessment of data. The search strategy in the various databases examined is reported in Table 1.

Inclusion and exclusion criteria

Inclusion criteria: Studies that reported the prevalence of *Toxoplasma gondii* in pregnant women. Studies for

Table 1 Search strategy in various databases studied

Database	Search Type	Search strategy
PubMed	Advanced search	((("prevalence"[Title/Abstract]) OR ("outcome"[Title/Abstract])) AND ("factors"[Title/Abstract])) AND ("toxoplasma gondii"[Title/Abstract]) AND ("pregnant women"[Title/Abstract])
ScienceDirect	Advanced search	Titke, abstract, keywords: (prevalence OR outcome) AND (factors) AND ("toxoplasma gondii") AND ("pregnant women")
Scopus	basic search	Titke, abstract, keywords: (prevalence OR outcome) AND (factors) AND ("toxoplasma gondii") AND ("pregnant women")
WOS	basic search	TS=(prevalence OR outcome) AND (factors) AND ("toxoplasma gondii") AND ("pregnant women" OR pregnancy)
Embase	Advanced search	prevalence:ab,ti AND 'toxoplasma gondii':ab,ti AND 'pregnant women':ab,ti
Google scholar	Advanced search	prevalence "pregnant women" OR pregnancy "toxoplasma gondii"

which the full text was available. Studies that provided sufficient data (sample size, prevalence percentage). Studies whose text was in English. Analytical observational studies (cross-sectional). Studies conducted in countries around the world and on different continents all *Toxoplasma gondii* diagnostic methods.

Exclusion criteria were reviews studies, intervention studies, cohorts, case-controls, duplicate papers, non-English investigations, and all articles with insufficient or unextractable data.

Study selection and quality assessment

In this process, initial searching was applied and selected papers were transferred to the Citation Management Software (EndNote). Duplicate articles were detected and excluded. Following the primary screening, the "Title" and "Abstract" of all studies were evaluated and un-eligible studies were excluded. In the second screening, full-text of remaining papers was examined and the articles with insufficient or unextractable data were ignored. Selected papers were enrolled for quality control step. The Newcastle–Ottawa Scale (NOS), a quality assessment tool for observational studies, was used to assess the quality of the articles reviewed in this study. The NOS assigns a maximum of 9 points for the lowest risk of bias for three areas examined in the articles: four points for the selection of study groups, two points for the comparison of groups, and three points for the determination of exposure and outcomes for case–control and cohort studies, respectively. Finally, in this review, articles were considered as high quality (score ≥ 5) or low quality (score less than 5). Finally, 138 articles were reviewed and analyzed in this study.

Data extraction and meta-analysis

Data extraction was applied using a previously prepared checklist including Name of first author, Year of

publication, Type of study, Study location, Sample size, Age range, Mean age, Number (or percentage of seroprevalence in 4 ranges of total, first, second, and third trimesters), and Number (or percentage) of IgG, IgM, and IgG/IgM antibodies (Table 2). After statistical extraction of data, Comprehensive Meta-Analysis software (CMA, v2) was hired for data analysis. Heterogeneity of the studies was assessed through the I^2 index and the Egger test was used to check publication bias followed by the Funnel Plot diagram. The random effects model was used in case of I^2 index above 50%. In order to investigate the factors affecting the heterogeneity of studies, meta-regression tests were used to examine factors such as sample size and year of study. Also, given the geographical scope of the study and the studies conducted in different countries around the world, subgroup analysis was also conducted based on continent.

Results

Following initial searching using valid databases and manual reviewing (Checking references of peer-reviewed articles), 1443 (PubMed (125), ScienceDirect (28), Scopus (308), Web of Sciences (WoS) (515), Embase (367), and Google Scholar (100) and 15 articles were selected in other resources, respectively (n:1458 in total). 898 duplicate articles were merged using EndNote software. During primary and secondary screenings, 560 and 411 studies were excluded, respectively and 11 low-quality articles were ignored. Finally, 138 eligible studies were included for data extraction (Fig. 1 and Table 2).

Meta-analysis

In the review of 138 studies with a sample size of 135,098 pregnant women individuals, heterogeneity index was found high ($I^2:98.9$); thus, random effect model was used for data analysis. The overall seroprevalence of *Toxoplasma gondii* in pregnant women was found 36.6% (95%CI:33.7–39.6) (Fig. 2). Also, Egger test showed no publication bias among the studies ($p:0.088$) (Fig. 3).

Table 2 Summary of the seroprevalence studies included in the review

Author	Year	Study	Region	Sample size	Age	Seroprevalence, n(%)				SP trimesters, n (%)			Total	SM	
						Range	Mean	IgG	IgM	IgG/ IgM	1st	2nd			3rd
Abamecha et al [23]	2016	Cross sectional	Ethiopia	232	-	-	23.65	191 (82.3)	0 (0)	7 (3)	56 (28.3)	100 (50.5)	42 (21.2)	198 (85.3)	ELISA
Abdelbaset et al [6]	2020	Cross sectional	Egypt	96	-	-	26.3	-	-	-	0 (0)	10 (35.7)	12 (23)	22 (22.9)	LAT/ELISA
Addo et al [1]	2023	Cross sectional	Ghana	84	-	-	26.96	44 (52.38)	3 (3.57)	-	11 (45.83)	17 (58.62)	19 (61.29)	47 (55.95)	ELISA
Agmas et al [24]	2015	Cross sectional	Ethiopia	263	18-44	-	29.56	-	-	-	-	-	-	180 (68.4)	LAT
Ahmadi et al [25]	2023	Retrospective crosssectional	Iran	4225	15-45	-	-	885 (20.8)	405 (9.5)	510 (12)	-	-	-	1800 (42.6)	ELISA
Ahmadpour et al [12]	2019	Cross sectional	Iran	276	-	-	33.1	55 (19.92)	6 (2.17)	5 (1.81)	-	-	-	66 (23.9)	ELISA CLIA
Ahmed et al [26]	2014	Cross sectional	Egypt	100	21-35	-	26.95	63 (63)	11 (11)	8 (8)	-	-	-	82 (82)	IHA
Akinbami et al [27]	2010	Cross sectional	Nigeria	179	25-30	-	-	73 (40.8)	-	-	-	-	-	73 (40.8)	ELISA
Akpınar et al [28]	2017	Retrospective crosssectional	Turkey	3140	-	-	-	344 (28.4)	34 (1.8)	-	378 (12)	0 (0)	0 (0)	378 (12)	Macro ELISA
Al Mohammad et al [29]	2010	Cross sectional	SAR	554	18-41	-	24.1	285 (51.4)	49 (8.8)	11 (2)	-	-	-	345 (62.3)	ELISA
Al-Adhroey et al [16]	2019	Cross sectional	Yemen	420	16-45	-	-	54 (12.9)	5 (1.2)	30 (7.1)	16 (18.8)	32 (21.3)	41 (22.2)	89 (21.2)	ECL / ELISA
Al-Eryani et al [30]	2016	Cross sectional	Yemen	593	15-48	-	-	259 (43.7)	54 (9.1)	44 (7.4)	-	-	-	269 (45.4)	ELISA
Al-Harathi et al [31]	2006	Cross sectional	SAR	197	17-45	-	-	58 (29.4)	11 (5.6)	4 (2)	64 (37.8)	9 (42.8)	0 (0)	73 (37)	ELISA
Alghamdi et al [32]	2016	Cross sectional	SAR	203	15-45	-	32	66 (32.5)	13 (6.4)	-	-	-	-	79 (38.9)	ELISA
Almogren [33]	2011	Retrospective crosssectional	SAR	2176	-	-	25	825 (38)	-	-	-	-	-	825 (38)	IHA
Alvarado et al [34]	2006	Cross sectional	Mexico	343	13 ≤	-	-	21 (6.1)	0 (0)	-	-	-	-	21 (6.1)	ELISA
Amar et al [35]	2015	Cross sectional	India	103	-	-	-	26 (25.2)	2 (1.9)	-	4 (21)	16 (27)	8 (32)	28 (27.18)	ELISA
Andiappan et al [36]	2014	Prospective crosssectional	Thailand	760	14-47	-	29.5	167 (22)	-	23 (3)	162 (26.1)	27 (20.9)	1 (10)	190 (25)	ELISA
Andiappan et al [37]	2014	Prospective crosssectional	Malaysia	219	-	-	-	93 (42.4)	-	-	3 (60)	32 (40.5)	58 (43)	93 (42.4)	ELISA
Andiappan et al [37]	2014	Prospective crosssectional	Myanmar	215	-	-	-	66 (30.7)	-	-	4 (66.7)	27 (28.4)	35 (30.7)	66 (30.7)	ELISA
Antinarelli et al [38]	2021	Retrospective crosssectional	Brazil	5895	-	-	-	2591 (44.4)	5 (0.1)	110 (1.9)	-	-	-	27.06 (45.9)	CMIA

Table 2 (continued)

Author	Year	Study	Region	Sample size	Age	Seroprevalence, n(%)				SP trimesters, n (%)			Total	SM	
						Range	Mean	IgG	IgM	IgG/IgM	1st	2nd			3rd
Asthana et al [39]	2006	Cross sectional	Grenada	534	-	-	-	304 (57)	-	-	-	-	-	304 (57)	ELISA
Atakorah et al [40]	2022	Cross sectional	Ghana	150	18 - 40	27.83	72 (48)	17 (11.3)	11 (7.3)	-	-	-	-	100 (66.66)	ELISA
Awoke et al [41]	2015	Cross sectional	Ethiopia	384	-	26.96	71 (18.5)	-	-	-	7 (17.9)	28 (20.6)	36 (17.2)	71 (18.5)	LAT
Ayeh et al [4]	2022	Cross sectional	Cameroon	300	16 - 41	28.05	218 (72.7)	4 (1.3)	18 (6)	-	-	-	-	240 (80)	ELISA
Aynioglu et al [42]	2015	Retrospective crosssectional	Turkey	910	17 - 46	29.13	400 (43.9)	23 (2.5)	-	-	-	-	-	423 (46.48)	CLIA
Baghel et al [43]	2020	Cross sectional	India	200	35 ≤	-	11 (5.5)	0 (0)	-	-	-	-	-	11 (5.5)	ELISA
Bamba et al [5]	2017	Cross sectional	Burkina Faso	316	-	26.9	98 (31.1)	-	-	-	-	-	-	98 (31.1)	ELISA/ ELFA
Barbosa et al [44]	2009	Cross sectional	Brazil	190	13 - 45	-	126 (66.3)	1 (0.52)	-	-	16 (12.6)	53 (41.7)	58 (45.7)	127 (66.8)	ELISA
Bashour et al [13]	2024	Cross sectional	Iran	340	16 - 46	29.13	101 (29.7)	-	2 (0.6)	-	-	-	-	103 (30.3)	ELISA/ PCR
Barzgar et al [2]	2024	Cross sectional	Iran	1200	18 - 42	-	381 (31.7)	41 (3.4)	-	-	-	-	-	422 (35.1)	ELISA
Bassiony et al [45]	2016	Cross sectional	Egypt	382	17 - 44	26.71	219 (57.33)	43 (11.3)	41 (10.7)	-	77 (58.8)	98 (55)	46 (63)	221 (57.9)	ELISA
Bjerke et al [46]	2011	Cross sectional	Pakistani immigrants in Norway	206	18 - 44	27.3	36 (17.4)	-	-	-	-	-	-	36 (17.4)	EIA
Chandrasena et al [47]	2016	Cross sectional	Sri Lanka	293	-	27	36 (12.3)	0	-	-	-	-	-	36 (12.3)	RDT
Chemoh et al [9]	2019	Cross sectional	Malaysia	219	20 - 40 <	-	67 (30.6)	5 (2.3)	4 (1.8)	-	4 (23.5)	38 (33.3)	34 (38.6)	76 (34.7)	ELISA
Costa et al [48]	2018	Cross sectional	Brazil	463	13 - 44	24	335 (72.3)	17 (3.7)	-	-	-	-	-	352 (76.02)	ELISA
Cvetković et al [49]	2010	Retrospective Crosssectional	Macedonia	235	15 - 45	-	48 (20.4)	-	-	-	-	-	-	48 (20.4)	ELISA
da Rocha et al [50]	2015	Cross sectional	Brazil	338	-	-	240 (71)	0	0 (0)	-	-	-	-	240 (71)	MEIA
Daka et al [11]	2024	Cross sectional	Zambia	401	24 - 30	27	17 (4.2)	3 (0.7)	1 (0.2)	-	0 (0)	11 (57.9)	8 (42.1)	19 (4.7)	ELISA
De Paschale et al [51]	2008	Cross sectional	Italy	3426	15 - 44	-	737 (21.5)	42 (1.2)	-	-	-	-	-	779 (22.7)	ELISA
De Paschale et al [52]	2014	Cross sectional	Benin	283	15 - 41	26.2	737 (21.5)	42 (1.2)	-	-	-	-	-	86 (30)	ELISA
De Quadros et al [53]	2015	Cross sectional	Brazil	148	14 - 41	-	24 (16)	1 (0.6)	-	-	-	-	-	25 (16.9)	-
Deka et al [54]	2022	Cross sectional	India	165	-	27.6	63 (38.2)	22 (13.3)	18 (10.9)	-	-	-	-	68 (41.2)	ELISA

Table 2 (continued)

Author	Year	Study	Region	Sample size	Age		Seroprevalence, n (%)			SP trimesters, n (%)			Total	SM
					Range	Mean	IgG	IgM	IgG/ IgM	1st	2nd	3rd		
Duarte et al [55]	2020	Cross sectional	Brazil	201	-	27.5	54 (26.8)	0	-	-	-	-	54 (26.8)	IFA
Dwinata et al [56]	2017	Cross sectional	Indonesia	330	17 - 40	27	36 (10.9)	-	-	-	-	-	36 (10.9)	ELISA
Edrees et al [18]	2020	Cross sectional	Iraq	150	-	29.9	40 (26.7)	0	-	9 (16.9)	19 (28.3)	12 (40)	40 (26.7)	ELISA
El Deeb et al [57]	2012	Cross sectional	Egypt	323	19 - 39	-	218 (67.5)	9 (2.8)	9 (2.8)	80 (69.6)	46 (63.9)	92 (67.6)	218 (67.5)	ELISA/VIDAS
Elsheikha [58]	2008	Cross sectional	Egypt	350	-	25.2	211 (60.3)	-	-	-	-	-	211 (60.3)	ELISA
Endris et al [59]	2014	Cross sectional	Ethiopia	385	20 - 30	-	341 (88.6)	-	-	14 (70)	228 (89.8)	99 (89.2)	341 (88.6)	LAT
Eshratkha et al [60]	2018	Cross sectional	Iran	620	14 - 47	-	114 (18.4)	-	3 (0.5)	117 (18.9)	0	0	117 (18.9)	ELISA
Fenta [61]	2019	Cross sectional	Ethiopia	494	15 - 42	26.47	370 (74.9)	19 (3.8)	15 (3.1)	79 (19.6)	140 (34.7)	185 (45.8)	404 (81.8)	ELISA
Findal et al [62]	2015	Cross sectional	Norway	1922	15 - 46	-	179 (9.3)	-	-	-	-	-	179 (9.3)	ELISA
Flores et al [63]	2021	Cross sectional	Panama	2326	-	-	997 (42.9)	5 (0.21)	31 (1.33)	-	-	-	1033 (44.4)	avidity test
Fonseca et al [64]	2012	Cross sectional	Brazil	2136	20 - 49	-	1057 (49.5)	77 (3.6)	-	-	-	-	1134 (53)	-
Frimpong et al [65]	2017	Cross sectional	Zambia	411	15 - 49	-	24 (5.87)	0	-	3 (3.57)	11 (8.21)	10 (5.18)	24 (5.87)	RDT
Garedaghi et al [66]	2017	Cross sectional	Iran	200	-	-	79 (39.5)	5 (2.5)	2 (1)	40 (43.4)	11 (44)	33 (39.7)	82 (41)	ELISA
Gelaye et al [67]	2015	Cross sectional	Ethiopia	288	18 - 42	28.41	-	-	-	14 (77.8)	52 (88.1)	180 (85.3)	246 (85.4)	LAT
Gheslaghi et al [15]	2022	Cross sectional	Iran	740	14 - 48	28.1	157 (21.2)	2 (0.27)	-	-	-	-	159 (21.5)	ELISA
Hajsoleimani et al [68]	2012	Cross sectional	Iran	500	-	-	186 (37.2)	7 (1.4)	-	74 (43.5)	52 (35.9)	63 (34)	189 (37.8)	ELISA
Hansu et al [69]	2021	Retrospective crosssectional	Turkey	29424	15 - 45	-	6035 (20.5)	502 (1.7)	255 (0.86)	6282 (21.34)	0	0	6282 (21.34)	-
Hariri et al [70]	2023	Cross sectional	Iran	244	16 - 43	23.06	54 (22.1)	0 (0)	-	-	-	-	54 (22.1)	ELISA
Hassanain et al [71]	2018	Cross sectional	Egypt	388	18 - 44	27	-	-	-	0 (0)	40 (33.1)	39 (35.8)	79 (20.4)	ELISA
Hosseini et al [72]	2023	Cross sectional	Iran	100	-	-	68 (68)	1 (1)	-	-	-	-	69 (69)	ELISA
Hung et al [73]	2007	Cross sectional	Sao Tome and Principe	499	-	25.4	-	-	-	-	-	-	375 (75.2)	LAT
Hung et al [74]	2015	Cross sectional	Taiwan	104	20 - 42	30.91	7 (6.7)	0	1 (1)	-	-	-	8 (7.7)	ELISA
Ijaz et al [75]	2020	Prospective Crosssectional	Pakistan	418	-	33	140 (33.5)	28 (6.7)	12 (2.9)	-	-	-	180 (43)	ELISA

Table 2 (continued)

Author	Year	Study	Region	Sample size		Seroprevalence, n (%)			SP trimesters, n (%)			Total	SM
				Age	Range	IgG	IgM	IgG/IgM	1st	2nd	3rd		
Iqbal et al [76]	2007	Cross sectional	Kuwait	19-41	27	119 (53.1)	31 (13.8)	-	150 (66.96)	0 (0)	0 (0)	150 (66.96)	ELFA/VIDAS
Jahantigh et al [77]	2020	Cross sectional	Iran	-	-	13 (14)	0 (0)	-	6 (15)	7 (14)	0 (0)	13 (14)	ELISA
Jiang et al [78]	2018	Cross sectional	China	-	-	43 (13.74)	4 (1.28)	4 (1.28)	-	-	-	51 (16.29)	ELISA
Jula et al [79]	2018	Prospective cross sectional	Ethiopia	-	23.1	-	-	-	19 (25.3)	42 (26.6)	35 (20.8)	96 (23.9)	EIA
Karabulut et al [80]	2011	Cross sectional	Turkey	18-40	29.8	408 (37)	15 (1.4)	97 (8.8)	520 (47.2)	0 (0)	0 (0)	520 (47.2)	Vitros ECIQ
Kassie et al [81]	2024	Cross sectional	Ethiopia	18-44	-	132 (38.8)	-	-	-	-	-	132 (38.8)	ELISA
Kaur et al [82]	1999	Prospective cross sectional	India	16-36<	-	-	14 (11.6)	-	-	-	-	14 (11.6)	-
Khademi et al [83]	2019	Cross sectional	Iran	14-55	27	100 (27.8)	-	3 (0.83)	-	-	-	103 (28.6)	ELISA
Khames et al [84]	2020	Cross sectional	Algeria	-	-	187 (18.5)	16 (1.6)	49 (4.8)	-	-	-	252 (25)	ELISA
Kiedmanee et al [85]	2019	Cross sectional	Thailand	15-49	-	-	-	-	-	-	-	33 (31.4)	ELISA
Linguissi et al [86]	2012	Retrospective crosssectional	Burkina Faso	18≤	-	37 (20.3)	7 (3.8)	-	44 (24.17)	0 (0)	0 (0)	44 (24.17)	ELISA
Liu et al [87]	2009	Cross sectional	China	21-38	-	25 (10.6)	0 (0)	-	-	-	-	25 (10.6)	ELISA
Lobo et al [20]	2017	Cross sectional	Portugal	15-44	-	17 (10.95)	0 (0)	17 (10.95)	-	-	-	34 (21.9)	DAT/ISAGA
Lobo et al [20]	2017	Cross sectional	Angola	15-44	-	71 (23.7)	6 (2)	5 (1.7)	18 (26.9)	41 (27.5)	23 (27.4)	82 (27.3)	DAT/ISAGA
Lopes-Mori et al [88]	2013	Cross sectional	Brazil	-	25	1122 (50.4)	-	-	-	-	93 (55)	1151 (51.7)	ELISA
Mahdy et al [89]	2017	Cross sectional	Yemen	22-30	26	154 (42.89)	7 (1.9)	-	-	-	55 (46.2)	166 (46.2)	ELISA
Majid et al [90]	2016	Cross sectional	Pakistan	15-40	-	135 (18.41)	-	-	21 (7.17)	43 (20.19)	71 (31.28)	135 (18.41)	IFA
Marković et al [91]	2023	Cross sectional	Serbia	15≤	-	-	0 (0)	-	13 (34.2)	6 (15.8)	19 (50)	38 (12.7)	VIDAS
Mohamed et al [92]	2016	Cross sectional	SAR	16-40	30.19	69 (21.2)	4 (1.2)	-	18 (25.7)	20 (23.5)	35 (20.8)	73 (22.4)	ELISA
Mosawi et al [93]	2019	Cross sectional	Afghanistan	-	-	-	-	-	-	-	-	207 (48.03)	CLIA
Moroi et al [94]	2020	Cross sectional	Romania	18-36<	-	2746 (39.86)	-	-	-	-	-	2746 (39.86)	CLIA
Mulugeta et al [10]	2020	Cross sectional	Ethiopia	18-42	26.11	-	-	-	27 (79.4)	59 (64.1)	72 (67.3)	158 (67.8)	LAT
Murad et al [95]	2023	Cross sectional	Iraq	15-50	-	110 (61.1)	-	-	33 (76.7)	27 (55.1)	50 (56.8)	110 (61.1)	ELISA

Table 2 (continued)

Author	Year	Study	Region	Sample size	Age		Seroprevalence, n(%)			SP trimesters, n (%)			Total	SM
					Range	Mean	IgG	IgM	IgG/ IgM	1st	2nd	3rd		
Murad et al [96]	2023	Cross sectional	Iraq	400	15- 50	-	205 (51.3)	-	-	-	-	-	205 (51.3)	ELISA
Mwambe et al [97]	2013	Cross sectional	Tanzania	350	15- 44	-	-	-	-	4 (26.7)	50 (33.3)	55 (29.7)	108 (30.9)	ELISA
Nasir et al [98]	2015	Cross sectional	Nigeria	360	18- 40	30	144 (40)	32 (8.9)	6 (1.7)	38 (-)	44 (-)	94 (-)	176 (48.88)	ELISA
Ndumbe et al [99]	1992	Cross sectional	Cameroon	192	-	-	148 (77.1)	-	-	-	-	-	148 (77.1)	EIA
Nijem et al [100]	2009	Cross sectional	Palestine	204	16- 43	26.4	57 (27.9)	36 (17.6)	-	93 (45.5)	0 (0)	0 (0)	93 (45.5)	ELISA
Nissapatorn et al [101]	2003	Cross sectional	Malaysia	200	18- 43	29.6	78 (39)	8 (4)	12 (6)	2 (33.3)	19 (45.2)	77 (49)	98 (49)	ELISA
Njunda et al [102]	2011	Cross sectional	Cameroon	110	18- 41	-	69 (62.7)	0 (0)	3 (2.7)	30 (75)	40 (60.6)	2 (50)	72 (65.4)	ELISA
Njunda et al [103]	2011	Cross sectional	Cameroon	110	20- 44	-	77 (70)	3 (2.7)	-	14 (87.5)	39 (66.1)	24 (70)	77 (70)	ELISA
Nowakowska et al [104]	2006	Cross sectional	Poland	4916	19- 46	26.7	2030 (41.3)	-	-	-	-	-	2030 (41.3)	IDAT/ ELISA
Ocak et al [105]	2007	Cross sectional	Turkey	1652	16- 42	27.2	860 (52.1)	9 (0.54)	-	-	-	-	869 (52.6)	ELISA
Olariu et al [106]	2020	Cross sectional	Romania	208	Dec-41	27.1	-	-	-	-	-	-	116 (55.8)	Vitros ECIQ
Oliveira et al [107]	2019	Cross sectional	Brazil	196	-	-	133 (67.9)	3 (1.5)	-	-	-	-	136 (69.3)	CLIA
Ouermi et al [108]	2009	Cross sectional	Burkina Faso	276	19- 42	27.65	75 (27.2)	13 (4.7)	-	-	-	-	88 (31.9)	ELISA
Oyinloye et al [109]	2014	Cross sectional	Nigeria	90	15- 35<	-	20 (22.2)	-	-	2 (18.1)	8 (29.6)	10 (19.4)	20 (22.2)	ELISA
Pereira et al [110]	2024	Cross sectional	Brazil	280	15- 43	26	69 (24.6)	-	-	35 (27.13)	9 (16)	25 (26.3)	69 (24.6)	CLIA
Qamer et al [111]	2020	Cross sectional	SAR	306	16- 45	29.9	99 (32.4)	3 (1)	-	-	-	-	101 (33)	ECL
Rocha et al [112]	2015	Cross sectional	Brazil	338	-	-	240 (71)	0 (0)	-	-	-	-	240 (71)	MEIA
Rosso et al [113]	2008	Cross sectional	Colombia	955	14- 40	25.1	437 (45.8)	27 (2.8)	-	-	-	-	464 (48.5)	MEIA
Sakikawa et al [114]	2012	Cross sectional	Japan	4466	16- 46	27.4	-	-	-	-	-	-	459 (10.3)	LAa titer
Salih [115]	2010	Cross sectional	Iraq	260	15- 44	-	80 (30.76)	31 (11.92)	-	-	-	-	111 (42.7)	ELISA
Sandoval et al [116]	2020	Cross sectional	Mexico	311	18- 44	-	13 (4.2)	0 (0)	-	-	-	-	13 (4.2)	ELISA
Santos et al [117]	2017	Cross sectional	Brazil	200	-	-	125 (62.5)	-	-	-	-	-	125 (62.5)	IFA

Table 2 (continued)

Author	Year	Study	Region	Sample size	Age	Seroprevalence, n(%)				SP trimesters, n (%)			Total	SM	
						Range	Mean	IgG	IgM	IgG/ IgM	1st	2nd			3rd
Sen et al [118]	2012	Cross sectional	India	380	-	-	-	-	74 (19.4)	-	74 (19.4)	0 (0)	0 (0)	74 (19.4)	ELISA
Sharbatkhori et al [119]	2014	Cross sectional	Iran	555	-	-	-	221 (39.8)	19 (3.4)	8 (1.4)	97 (47.3)	76 (45.7)	67 (36.4)	232 (41.8)	ELISA
Silva-Díaz et al [120]	2020	Cross sectional	Peru	218	-	-	25.5	66 (30.3)	12 (5.5)	-	20 (34.4)	26 (38.8)	32 (34.4)	78 (35.8)	ELISA
Singh et al [121]	2021	Cross sectional	Ghana	400	-	-	-	229 (57.3)	14 (3.5)	12 (3)	231 (57.8)	0 (0)	0 (0)	231 (57.8)	ELISA
Sirin et al [122]	2017	Retrospective Cross sectional	Turkey	7513	18 - 45	-	-	2427 (32.3)	138 (1.9)	-	-	-	-	2565 (34.1)	CEIM
Sitoe et al [123]	2010	Cross sectional	Mozambique	150	15 - 50	32	28 (18.7)	1 (3.6)	-	-	29 (19.3)	0 (0)	0 (0)	29 (19.3)	-
Sroka et al [124]	2010	Cross sectional	Brazil	963	Dec-44	-	-	661 (68.6)	5 (0.5)	-	-	-	-	666 (69.1)	MEIA
Tamer et al [125]	2009	Retrospective Cross sectional	Turkey	1972	-	-	28.56	952 (48.3)	8 (0.4)	31 (1.6)	991 (50.2)	0 (0)	0 (0)	991 (50.2)	ELISA
Teweldemedhin et al [126]	2019	Cross sectional	Ethiopia	360	15 - 49	-	-	117 (32.5)	-	11 (3.1)	41 (36)	42 (32.3)	45 (38.8)	128 (35.5)	ELISA
Uysal et al [127]	2013	Retrospective crosssectional	Turkey	4691	15 - 45	-	-	1871 (39.9)	0 (0)	48 (1.02)	1919 (41)	0 (0)	0 (0)	1919 (41)	ELISA/VIDAS
van Enter et al [128]	2017	Retrospective crosssectional	Thailand	199	16 - 46	26	-	-	-	-	-	-	-	63 (31.7)	ELISA
Vueba et al [129]	2020	Cross sectional	Angola	878	15 - 47	29.7	346 (39.4)	0 (0)	0 (0)	2 (0.2)	97 (35.2)	148 (39.1)	103 (45.7)	348 (39.6)	ECL
Yasmeen et al [130]	2017	Cross sectional	India	251	18 - 35	-	-	53 (21.1)	-	-	-	-	-	53 (21.1)	ELISA
Zemene et al [131]	2012	Cross sectional	Ethiopia	201	17 - 35	23.64	163 (81.1)	2 (1)	2 (1)	3 (1.5)	23 (79.3)	86 (82.7)	59 (86.8)	168 (83.6)	ELISA
Adeniyi et al [7]	2023	Cross sectional	Nigeria	320	14 - 50	-	-	76 (23.75)	35 (10.94)	-	-	-	-	111 (34.69)	RDT
ALANAZI et al [3]	2017	Prospective Crosssectional	Saudi Arabia	461	16 - 45	26.3	189 (40.9)	0 (0)	0 (0)	-	8 (21.6)	77 (37.3)	104 (47.7)	189 (40.99)	ELISA
Aleem et al [17]	2018	Cross sectional	Pakistan	360	18 - 33	-	-	-	-	-	71 (61.7)	59 (58.4)	40 (27.7)	170 (47.2)	LAT
Ballah et al [8]	2017	Prospective Crosssectional	Nigeria	400	15 - 49	-	-	-	-	-	38 (21.3)	57 (33.5)	17 (32.7)	112 (28)	ELISA
Cédric et al [22]	2022	Cross sectional	Cameroon	117	14 - 40	-	-	27 (23.1)	1 (0)	2 (1.7)	3 (20)	27 (30)	0 (0)	28 (24)	-
Dawet et al [14]	2022	Cross sectional	Nigeria	356	15 - 49	-	-	105 (29.5)	-	-	25 (27.78)	55 (26.57)	25 (42.37)	105 (29.5)	ELISA
Khan et al [21]	2018	Cross sectional	Pakistan	150	20 - 50<	-	-	1 (0.6)	1 (0.6)	-	-	-	-	2 (1.32)	RDT
Lushina et al [32]	2023	Cross sectional	Tanzania	383	-	-	-	102 (26.6)	2 (0.52)	-	25 (31.3)	40 (26)	39 (29.2)	104 (27.2)	-

Table 2 (continued)

Author	Year	Study	Region	Sample size	Age	Seroprevalence, n(%)			SP trimesters, n (%)			Total	SM	
						Range	Mean	IgG	IgM	IgG/ IgM	1st			2nd
Murebwayire et al [133]	2017	Cross sectional	Rwanda	384	18 - 45	27	37 (9.6)	15 (3.9)	5 (1.3)	8 (10)	26 (12.6)	13 (3.4)	47 (12.2)	ELISA
Nadia et al [134]	2023	Cross sectional	Cameroon	242	16 - 46	-	152 (62.8)	28 (11.6)	20 (8.3)	79 (75.2)	77 (80.2)	24 (58.5)	200 (82.7)	ELISA
Negero et al [135]	2017	Cross sectional	Ethiopia	210	15 - 44	-	-	-	-	20 (74)	62 (77.5)	77 (74.8)	159 (75.7)	LAT
Paul et al [136]	2018	Cross sectional	Tanzania	254	-	29.9	102 (40.2)	23 (9.1)	12 (4.7)	-	-	-	113 (44.5)	ELISA
Shah et al [137]	2017	Cross sectional	Pakistan	100	17 - 36	-	7 (7)	12 (12)	7 (7)	2 (6.45)	3 (8.82)	7 (20)	12 (12)	RDT
Song et al [138]	2005	Cross sectional	Korea	5725	20 - 40	-	-	-	-	-	-	-	51 (0.88)	ELISA/LAT

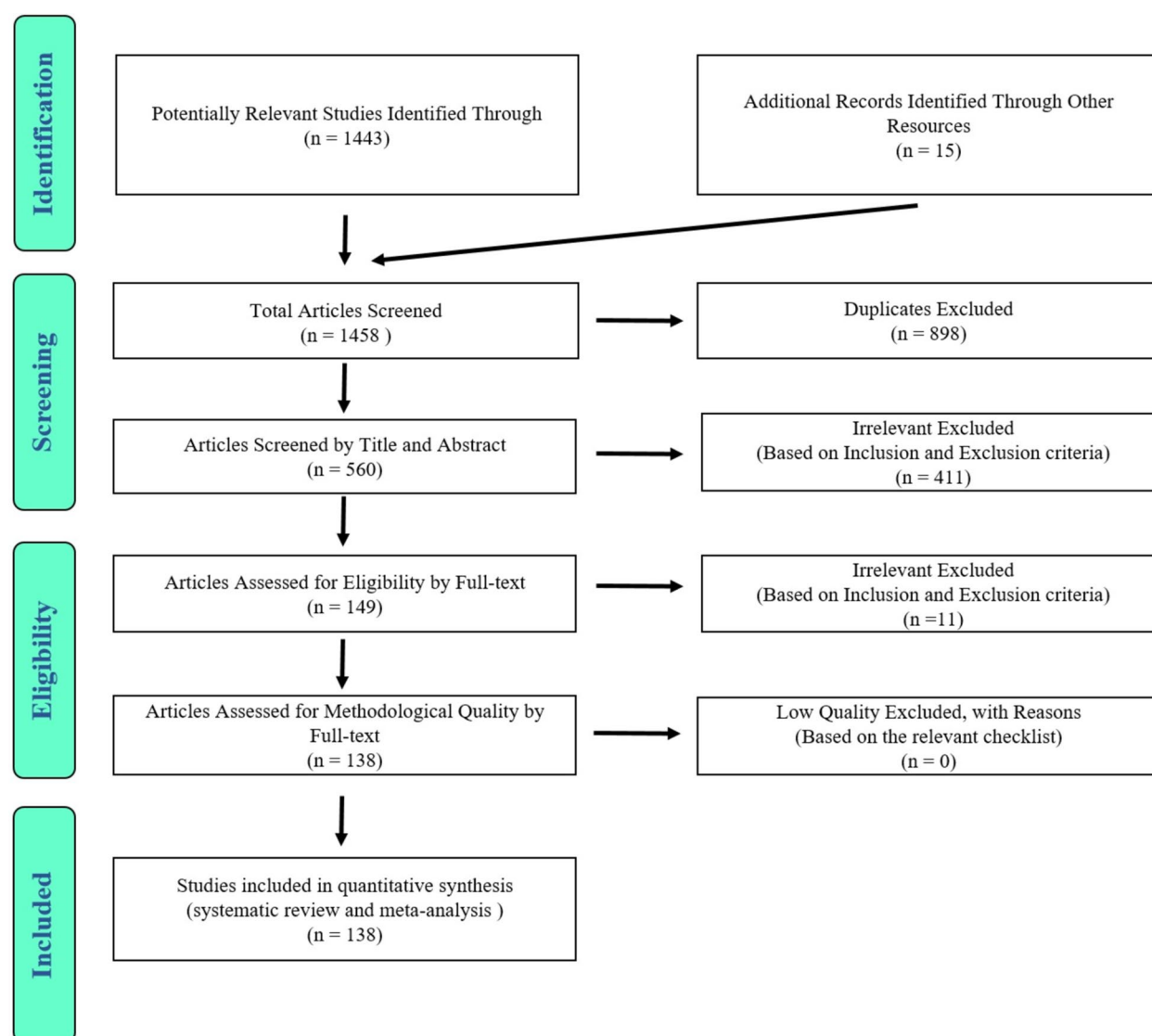


Fig. 1 Flowchart of PRISMA criteria (2020) indicating the stages of article selection

Meta-regression analysis

Meta-regression analysis showed that following the increase in sample size (Fig. 4) and year of paper publication (Fig. 5), the seroprevalence of *Toxoplasma gondii* in pregnant women decreased significantly ($p < 0.05$).

Subgroup analysis

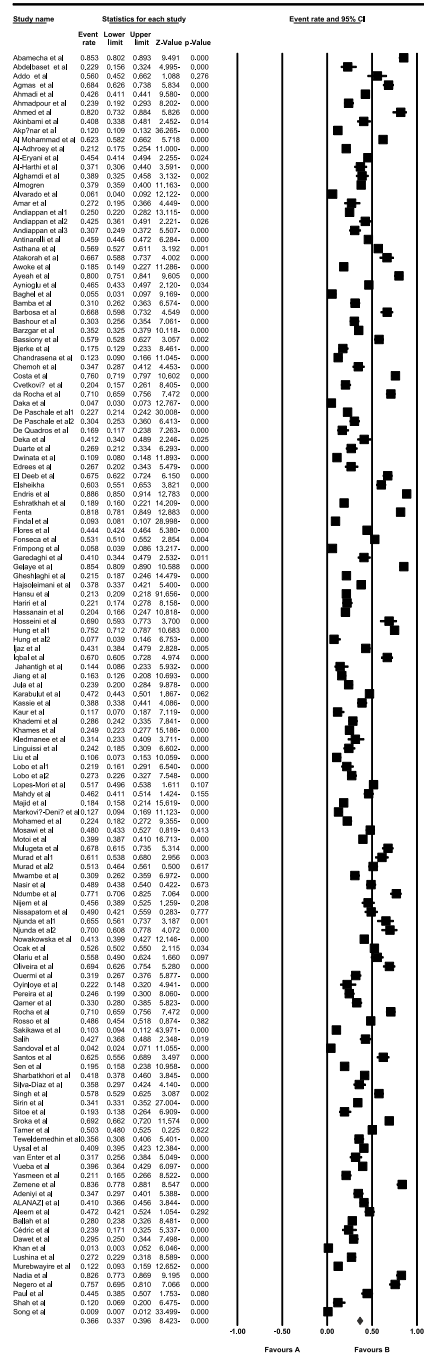
According to the results of Table 3, which reports the Global seroprevalence of *Toxoplasma gondii* in pregnant women by continent, the highest prevalence reported based on meta-analysis was reported in South America with 52.8% (95% CI:46.6–59), while only 15 studies were reviewed in this continent, most of which were in Brazil. Therefore, after the continent, the highest prevalence

reported was reported in Africa with 46.8% (95% CI:39.5–54.3). Also, the lowest prevalence reported based on meta-analysis was in North America with 19.7% (95% CI:8.4–39.6) and Europe with 24.6% (95% CI:17.8–32.9) (Table 3).

Discussion

This systematic review and meta-analysis study investigated the global seroprevalence of *Toxoplasma gondii* in pregnant women. In this regard, 138 eligible studies with the sample size of 135,098 pregnant women individuals were examined. The overall seroprevalence of *Toxoplasma gondii* in pregnant women was found 36.6%, the highest prevalence reported based on meta-analysis

Meta-analysis



Meta Analysis

Fig. 2 Forest plot of TP seroprevalence based on random effect model. TP: Toxoplasmosis in pregnant

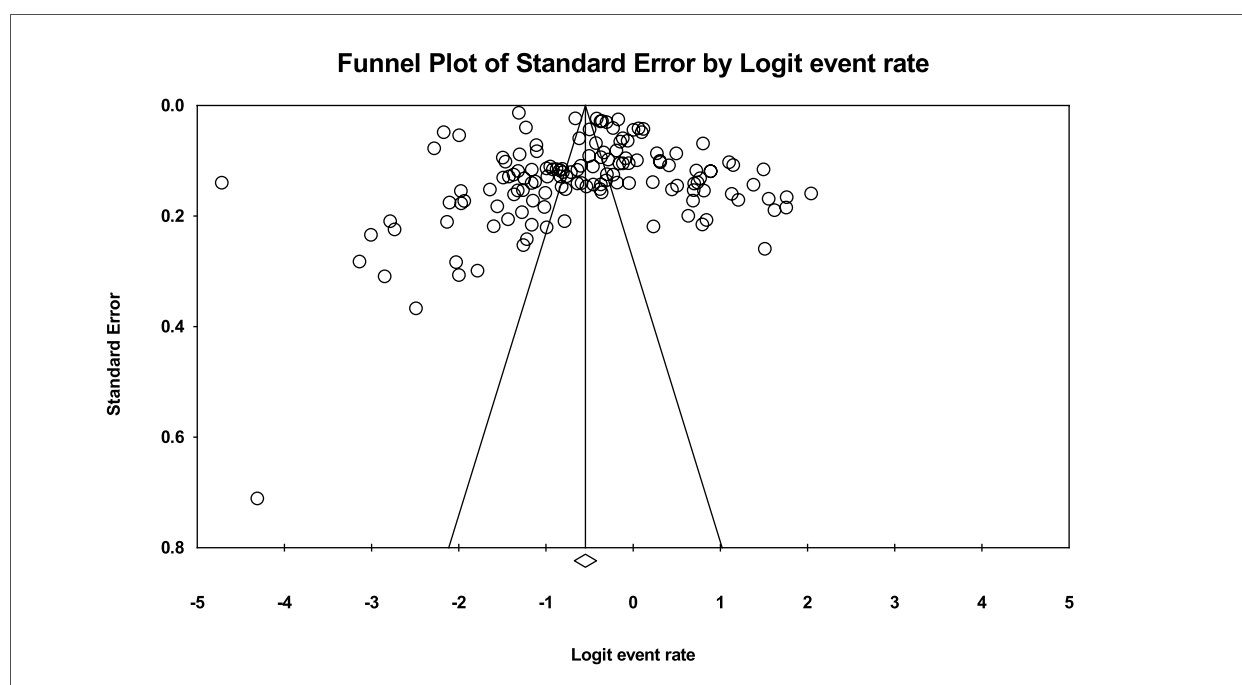


Fig. 3 Funnel plot representing distribution bias in selected eligible studies

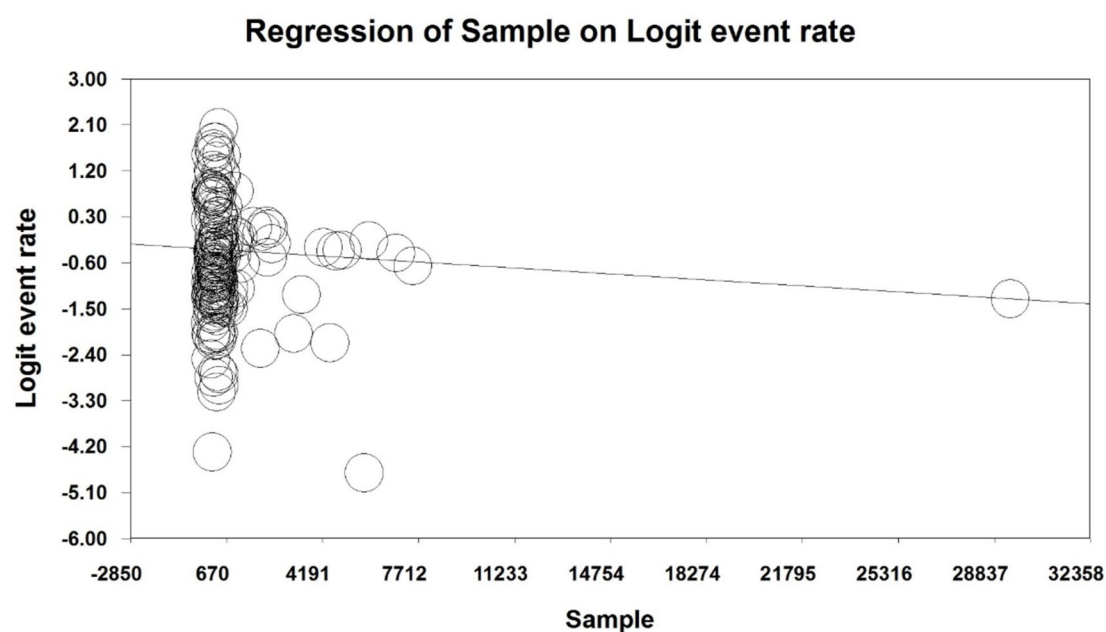


Fig. 4 Meta-regression represents the effect of sample size on the prevalence of Toxoplasma in pregnant women

was reported in South America with 52.8%, while only 15 studies were reviewed in this continent, most of which were in Brazil. Therefore, after the continent, the highest prevalence reported was reported in Africa with 46.8%. Also, the lowest prevalence reported based

on meta-analysis was in North America with 19.7% and Europe with 24.6%.

According to the researches, Toxoplasmosis in pregnancy are mostly diagnosed in the third trimester of pregnancy [1, 4, 8, 13, 15]. This fact returns to the

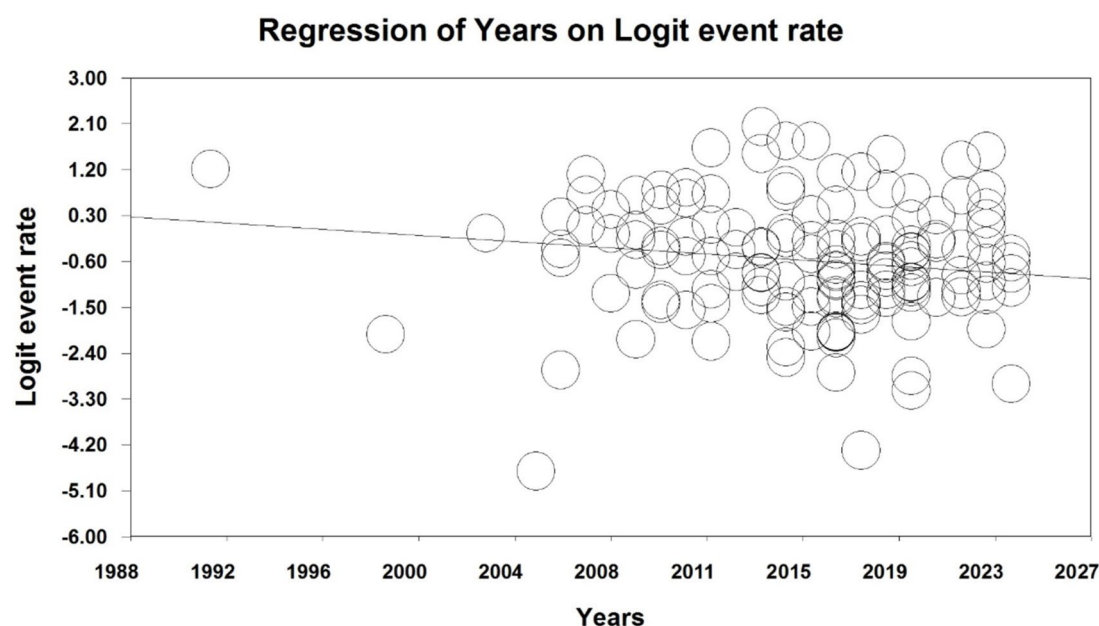


Fig. 5 Meta-regression represents the effect of the year of publication on the prevalence of Toxoplasma serum in pregnant women

Table 3 Global seroprevalence of *Toxoplasma gondii* in pregnant women by continent

Continent	N	Sample size	Heterogeneity (I^2)	Egger test	Prevalence (95% CI)
Africa	47	14,579	98.4	0.345	46.8 (95% CI:39.5–54.3)
Asia	64	84,001	98.9	0.385	29.4 (95% CI:25.9–33.1)
Europe	9	18,257	99.1	0.150	24.6 (95% CI:17.8–32.9)
North America	3	3514	98.8	0.256	19.7 (95% CI:8.4–39.6)
South America	15	14,747	97.8	0.408	52.8 (95% CI:46.6–59)

suppressive role of immune system in gestation, especially in third trimester [23].

However, the severity of fetal complications is higher in first months of pregnancy such as mental retardation, chorioretinitis, and blindness [3, 7, 13]. Thus, TG screening is strictly recommended for those exposed to the associated risk factors [3, 4]. The prevalence of TG infection depends on the presence of several risk factors such as living environment, education level, presence of cat in home, consumption of undercooked meat, and contact with contaminated soil. However, some other differences are associated with cultural beliefs and lifestyles [12, 17, 60, 71].

Most of the studies reported a considerable relationship between the prevalence of Toxoplasmosis and raw meat [2, 18, 31, 33, 61, 97, 134]. However, some studies reported no significant relationship in this regard [4, 23, 24, 67, 131]. These different reports also depend on the frequency of raw meat consumption and type of meat

(pig, sheep) [23, 131]. According to studies, the educational status of women represents a considerable relationship with the seroprevalence of TG. Thus, a lower educational level is considered a risk factor for onset of TG in pregnant women [2, 4, 21, 24, 33, 85, 134].

Most studies reported a significant relationship between higher age and accelerated prevalence of Toxoplasmosis [4, 24, 26, 28, 31, 61, 85, 97, 131, 134]. For example, Mwambe et al. reported that the risk of TG infection in pregnant women increases by 7% in each year of the age [97]. According to studies, tropical and humid weather are important risk factors in increasing the prevalence of Toxoplasmosis in pregnant women [2, 8, 23]. In such areas, the growth and development of TG eggs are environmentally probable [2]. It should be noted that each of the above-mentioned risk factors alone has no considerable effect on Toxoplasmosis onset, however simultaneous presence of several risk factors could induce a higher rate of *Toxoplasma gondii*.

The most important measures are the use and benefit of primary and secondary prevention, that is, firstly, contact and disease development can be prevented and in case of contact, screening and diagnosis of the disease can be increased [139]. The primary prevention strategy includes providing educational materials on possible preventive measures that should be integrated into prenatal visits, classes, and programs [139]. Among the primary preventive measures, women should avoid sources of infection, not drink unfiltered water, and observe hand hygiene. Screening as a secondary prevention is essential for early identification of infected women. However, until screening programs are implemented, it can be said that primary prevention and the use of educational programs are much more effective in reducing the incidence [139].

Some important limitations were detected in this systematic review and meta-analysis, including non-uniformity of sample size or data and the use of different serological methods with unique sensitivity and characteristics. It is suggested that future studies in this field should consider population prevalence, as well as review interventional and diagnostic studies in this area and provide their results to researchers.

Conclusion

The overall seroprevalence of *Toxoplasma gondii* in pregnant women was high at 38.6% globally. Therefore, substantial efforts are needed to educate pregnant women and to implement policy measures globally. Therefore, increasing public awareness about *Toxoplasma gondii* and its associated risk factors is essential. In addition, rigorous health programs should be considered by health policymakers for the prevention and early detection of *Toxoplasma gondii* in pregnant women. In this context, screenings at specific times for pregnant women can be considered.

Abbreviations

TP	Toxoplasmosis in pregnancy
SM	Serological method
ELISA	Enzyme-linked Immunosorbent assay
LAT	Latex agglutination test
CLIA	Chemiluminescence immunoassay
ECL	Electrochemiluminescence
IHA	Indirect hemagglutination test
CMIA	Chemiluminescent microparticle immunoassay
ELFA	Enzyme-linked fluorescence assay
PCR	Polymerase chain reaction
RDT	Rapid diagnostic test
EIA	Enzyme-linked immunoassay
MEIA	Microparticle enzyme-linked immunoassay
IFA	Indirect immunofluorescence assay
VIDAS	Vitek immunodiagnostic system
ISAGA	Immunosorbent agglutination assay
CEIM	Chemiluminescent enzyme immunoassay method

Acknowledgements

By Student Research Committee of Kermanshah University of Medical Sciences.

Authors' contributions

NS and AR and MM contributed to the design, MM statistical analysis, and participated in most of the study steps. MM and AR and HZ and SR prepared the manuscript. MM and HZ and SHS and AA assisted in designing the study, and helped in the interpretation of the study. All authors have read and approved the content of the manuscript.

Funding

Not applicable.

Data availability

Datasets are available through the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Received: 12 November 2024 Accepted: 15 January 2025

Published online: 30 January 2025

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